1. To transfer the power from the combustion chamber to the crank shaft through the connecting rod.
2. To transmit the heat produced from the power to the combustion chamber walls
3. To acts as seal inside the cylinder to withstand high pressure and protects the crankcase in avoiding high pressure to pass to the crank case
4. It acts as a guide to connecting rod
5. It acts as a component for supporting the piston ring
6. It is used to suck the air or air fuel mixture and also used to compress them
7. It is used to expel the hot gases to the atmosphere during the exhaust stroke.

5.3.7.3 Types of piston

Piston is used to transmit power from engine combustion chamber to the crankshaft. Pistons are classified according to their heads hased on the type of engine and its performance. They are classified as follows,

a. Flat head piston
b. Domed head piston
c. Concave (bowl) head piston

a) Flat head piston

In this type, the piston head is in flat shape. This shape helps in removing combusted products after the power stroke. The design of these pistons is very easy. However, the efficiency of the engines using this type of pistons is less.



Figure 5.3.7.3(a) Flat Piston

b) Domed head piston

In this type, the piston head is of cap like structure. ie., more volume is added to the piston head. This structure helps in smooth compression stroke and also results in proper mixing. Compression ratio is higher for using these types of piston. However the manufacturing process is difficult.

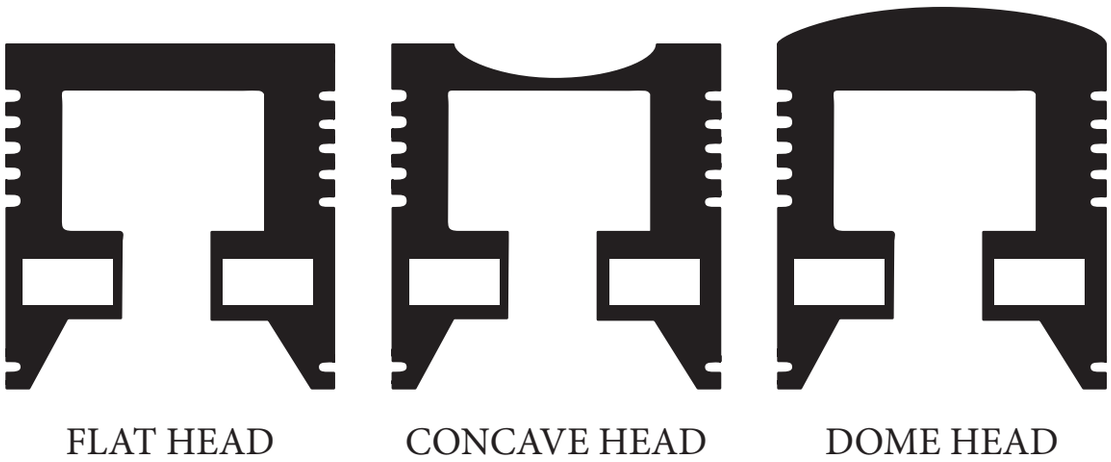


Figure 5.3.7.3 Types of Piston Head

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